

Lab 7 Worksheet — Network Devices and Traffic Behaviour Analysis

Name: ____ Date: ____

Exam objective: Distinguish hub, switch, router, firewall and access point by the layer they operate at and the traffic behaviour each produces (Core 1 objectives 2.2 and 2.3).

Record as you go

Step	What you did	What you observed
1	Open PCAP Analyzer at https://alfredang.github.io/pcapanalyzer/ and click the Sample button to generate a demonstration capture. Nothing is uploaded — parsing happens in your browser.	
2	Record the four dashboard metrics: packet count, total bytes, capture duration and average packet size.	
3	Open the protocol distribution view and list every protocol present with its share of the capture.	
4	Open Top talkers and record the three most active endpoints by traffic volume, noting their addresses.	
5	Open Top conversations and identify which pairs of hosts exchange the most data, then state what kind of session each pair most likely represents.	
6	Select an individual packet in the packets table and open its detail view to read the source, destination, protocol and length.	
7	From the packet detail, identify the Layer 2 MAC addresses and the Layer 3 IP addresses, and explain which one a switch uses to forward and	

Step	What you did	What you observed
	which one a router uses.	
8	Build the device comparison matrix with rows for hub, switch, router, firewall and access point and columns for OSI layer, forwarding basis, collision domains, broadcast domains and typical use.	
9	Explain in writing why a hub creates one collision domain across all ports while a switch creates one per port, and what that means for network performance.	
10	Explain why a switch forwards a broadcast frame out of every port but a router does not forward it at all, and connect this to why VLANs are needed.	
11	Apply the protocol filter in the packets table to isolate one protocol, and record how the packet count changes.	
12	Write a short conclusion stating what the capture reveals about the network — how many broadcast domains are visible, and whether a router is present in the path.	

Verification

Success criterion: Your matrix correctly assigns each of the five devices to its OSI layer and forwarding basis; your analysis names the specific MAC and IP addresses observed; and the conclusion is supported by evidence from the capture rather than from theory alone.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
